

Electrode Selection — Quick-Reference Card

Hands-On Electroretinography in the Age of AI · Chapter 2, Figure 2.1 / Tables 1 & 2 · For quick clinical reference only — see Chapter 2 for full rationale

Contact Lens	Gold Foil	DTL Fiber	Skin Electrode
1	2	3	4
Clinical setting Specialist laboratory vs. community or portable screening	Sensitivity requirement Which retinal layers / pathological changes must be detected	Ocular-contact tolerability Corneal disease, recent surgery, or high anxiety about eye contact	Skin-electrode tolerability Active periocular skin conditions or dermatitis

By Clinical Setting (Table 1)

Setting	Preferred Electrode	Reason
Specialist ERG laboratory	Contact lens or gold foil	Maximum amplitude, full ISCEV protocol, topical anesthesia available, trained operator
Secondary care outpatient clinic	Gold foil or DTL fiber	Good amplitude, no anesthesia needed, tolerated for full protocol duration
Community screening / portable clinic	DTL fiber or skin	Practical, disposable, patient-friendly, suitable for non-specialist operators
Pediatric clinic (cooperative child)	DTL fiber	No anesthesia, gentler than contact lens, sufficient amplitude for most protocols
Pediatric clinic (exam under anesthesia)	Contact lens	Anesthesia already administered; contact lens gives maximum data from single session
Telehealth / home-based monitoring (research)	Skin electrode	Self-applicable, safe, does not require clinical supervision

By Clinical Indication (Table 2)

Clinical Indication	Minimum Recommended Electrode
Suspected retinitis pigmentosa / rod dystrophy	Gold foil or DTL fiber
Suspected cone dystrophy	Gold foil or DTL fiber
Early diabetic retinopathy screening (OP analysis)	DTL fiber minimum; contact lens or gold foil preferred
Glaucoma detection (PhNR analysis)	Contact lens or gold foil — skin not recommended
Hydroxychloroquine toxicity monitoring	Gold foil or DTL fiber
Baseline check, low-resource / screening context	Skin electrode acceptable